



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 13	Team No- 21	Project Title- Student Attendance Management System
------------	-------------	---

1. Abstract

Manual attendance tracking in academic institutions is time-consuming, error-prone, and difficult to audit or scale across multiple courses and sections. This project proposes a Student Attendance Management System (SAMS) built as a polyglot, microservices-based backend to address these gaps end-to-end. The system uses a normalised PostgreSQL schema for core academic entities (students, courses, sections, enrolments) and a MongoDB document store for flexible, high-volume attendance logs, integrated through a dual-database consistency strategy. A FastAPI service exposes secure REST APIs for marking, querying, and reporting attendance, using JWT-based authentication with role-based access control for faculty, students, and administrators. Complementary services are implemented in Node.js/Express (real-time session handling via Socket.io) and Spring Boot (reporting and analytics with JPA/PostgreSQL), communicating through an API Gateway that centralises routing, token forwarding, and error handling. A pgvector-backed similarity search enables an optional face-embedding attendance verification mode. Each service owns its own database, following a database-per-service pattern, with timeouts and Saga-based compensating transactions handling cross-service failures. The system is containerised with Docker, deployable on Kubernetes, and monitored using OpenTelemetry, Prometheus, and Grafana, with a GitHub Actions pipeline automating testing and image scanning.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090119	A. Siddhartha Varma	
2520090202	P. Gowtham Varma	

3. Frameworks / Platforms / Software Used

Databases: PostgreSQL, MongoDB, pgvector. Backend: FastAPI, SQLAlchemy, Pydantic, Motor, Node.js/Express, Mongoose, Socket.io, Spring Boot, Spring Data JPA. Security: JWT, OAuth2, Spring Security. DevOps: Docker, Docker Compose, Kubernetes, GitHub Actions. Observability: OpenTelemetry, Prometheus, Grafana. Testing: pytest.

Guide Name and Signature (remarks if any) _____